

# Cmod A7-35T Power Specification

## 1. Overview

The Cmod A7-35T uses a **Linear Technologies LTC3569** triple-output buck regulator (IC10) to generate all onboard supply voltages from a single input rail called **VU**.

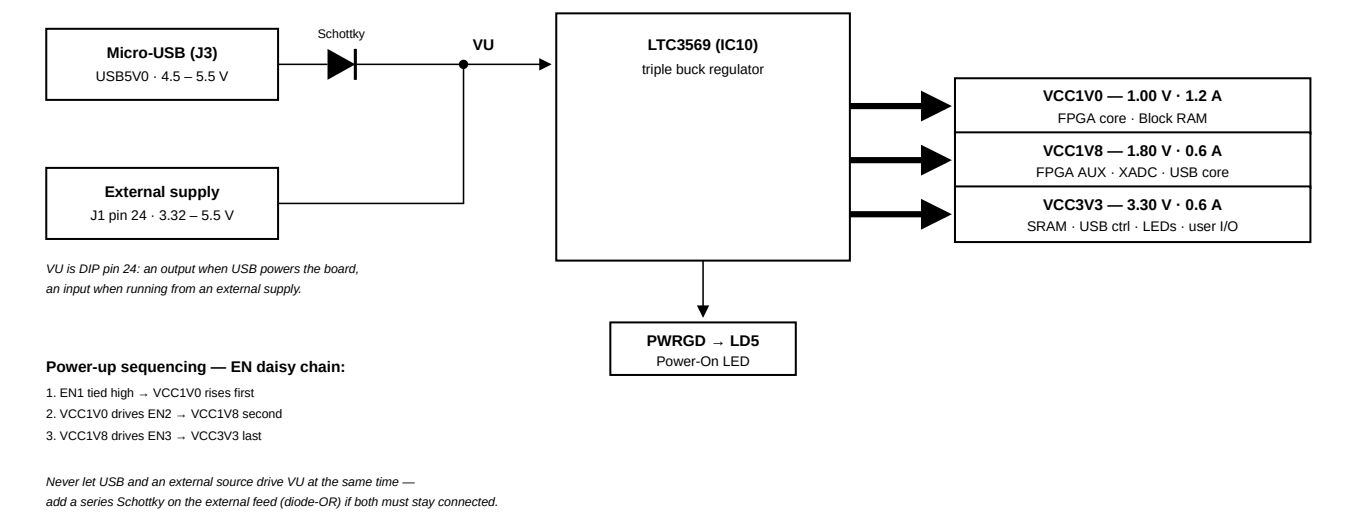


Figure 1. Cmod A7 power supply tree

## 2. Power Input Options

The board can be powered from either of the following sources:

| Source          | Connector       | Schematic Net | Min Voltage | Recommended | Max Voltage |
|-----------------|-----------------|---------------|-------------|-------------|-------------|
| USB             | J3 (Micro-USB)  | USB5V0        | 4.5 V       | 5.0 V       | 5.5 V       |
| External Supply | J1 Pin 24 (DIP) | VU            | 3.32 V*     | 5.0 V       | 5.5 V       |

\* The minimum external voltage on VU depends on the current drawn from the VCC3V3 rail via the Pmod header:

- 0 mA drawn → minimum 3.32 V
- 100 mA drawn → minimum 3.38 V
- 250 mA drawn → minimum 3.48 V

GND reference is on **J1 Pin 25**.

## 3. Output Power Rails

All three rails are generated by the LTC3569 from the VU input.

| Rail   | Typical Voltage | Min / Max Voltage | Max Current | Powered Devices                                          |
|--------|-----------------|-------------------|-------------|----------------------------------------------------------|
| VCC1V0 | 1.00 V          | 0.95 V / 1.05 V   | 1.2 A       | FPGA Core, Block RAM                                     |
| VCC1V8 | 1.80 V          | 1.71 V / 1.89 V   | 0.6 A       | FPGA AUX, ADC, USB Core                                  |
| VCC3V3 | 3.30 V          | 3.135 V / 3.465 V | 0.6 A       | SRAM, USB Controller, Pmod, LEDs, Buttons, FPGA User I/O |

## Power-Up Sequence

The LTC3569 enable pins are daisy-chained to enforce a sequential startup:

1. **VCC1V0** starts first (EN1 tied high)
2. **VCC1V8** starts after VCC1V0 is stable (EN2 driven by VCC1V0)
3. **VCC3V3** starts after VCC1V8 is stable (EN3 driven by VCC1V8)

The **PWRGD** (Power Good) output drives the Power On LED (LD5).

## 4. VU Pin Behavior

| Condition         | VU Pin Function                                                                                  |
|-------------------|--------------------------------------------------------------------------------------------------|
| USB connected     | VU is an <b>output</b> — driven by USB 5V through a Schottky diode (expect ~5 V with small drop) |
| USB not connected | VU is an <b>input</b> — accepts external power supply (3.32–5.5 V)                               |

## 5. Dual Power Source (USB + External)

The on-board Schottky diode is located on the **USB side** (between USB VBUS and VU), not on the VU input pin. This means:

| Scenario                                                            | Safe? | Notes                                                |
|---------------------------------------------------------------------|-------|------------------------------------------------------|
| USB only                                                            | ✓     | Normal operation                                     |
| External only (VU pin)                                              | ✓     | Normal operation                                     |
| Both connected, only one active                                     | ✓     | The inactive source supplies no current, no conflict |
| Both connected <b>and both supplying power</b>                      | ✗     | Two voltage sources fight on the same VU node        |
| Both connected and both active, <b>with external Schottky added</b> | ✓     | Higher-voltage source wins; lower is blocked         |

**Key distinction:** Physically connecting both cables is not the problem — the danger is having **both sources actively supplying voltage at the same time** without isolation.

If you need to have both connected simultaneously with both powered (e.g., USB for JTAG/UART while also on external supply), **add a Schottky diode in series with the external supply on the VU pin**. This creates a "diode-OR" circuit where whichever source has the higher voltage automatically takes over, and the other is safely blocked.

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## 6. Warnings

- When a USB host is attached, the VU pin is driven to USB voltage (4.5–5.5 V). Any external power source on VU **must be disconnected first** before plugging in USB, or the external supply may be damaged.
- This is **especially dangerous with battery power sources**, as batteries cannot tolerate reverse current.
- The FPGA I/O pins on the DIP connector have **no series resistors**. The Artix-7 absolute maximum input rating is  $V_{CCO} + 0.55\text{ V} = \mathbf{3.85\text{ V}}$  (and  $-0.4\text{ V}$  minimum, per DS181) — the pins are **not 5 V tolerant**; use a level shifter or divider for 5 V devices.

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## 7. Quick Reference for External Power Design

| Parameter                           | Value                            |
|-------------------------------------|----------------------------------|
| Input Connector                     | J1 Pin 24 (VU) + Pin 25 (GND)    |
| Recommended Input Voltage           | 5.0 V                            |
| Acceptable Input Range              | 3.32 – 5.5 V                     |
| Total Max Current (all rails)       | ~2.4 A (1.2 + 0.6 + 0.6)         |
| Power Good Indicator                | LD5 (LED)                        |
| Regulator IC                        | LTC3569                          |
| Protection Required for Dual Supply | Schottky diode in series with VU |